SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Total Marks: 20

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I A. Shanbagam / 192126060 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shanbagam. A.

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Cloud Computing Fundamental

* cloud computing provides computing resources through Internet

* Resources include Servers, Storage, networking and Software

* on-demand acess provides resources whenever required

* Resource pooling allows resourses to be Shared among uses

* Elasticity allows resures to increase of decreasing according to demand

* Scalability helps the System handle growing workloads.

* Measured Service allows uses to pay based on resource usage. Cloud computing reduces Infrastructure & maintance costs.

* It provides flexibility & global acess to resources & Improves resorce utilization.

Cloud Computing Fundamentals

Cloud Computing

Definition

A model for delivering computing resources (Servers, Storage, applications) over the Internet on a pay-as-you-go bases

Characteristics

- On demand access - Resources when are available when needed
- Resource pooling - Shared resources for multiple uses.
- Elasticity - Scale up or down based on demand
- Scalability - Handles growing workload efficiently
- Measured Usage

Benefits

- Cost reduction
- flexibility
- High availability
- Better resource utilization
- Global Access

Limitations

- Security Concerns
- Downtime risk
- Internet Dependency
- Less control
- Compliance Issues

Evolution Of Cloud Computing:

Cloud computing developed through the continuous evolution of computer hardware, Internet Software & distributed Computing Systems.

Hardware Evolution Improve processors, Servers & Storage. powerful hardware Increased overall Computing Capacity

Virtualization Improved resoured utilization. Virtualization allowed multiple Virtual Machines on one Physical Server. Distributed System Connected multiple Computer through networks. Distributed System Supported resource Sharing & processing

Internet Evolution Enabled global connectivity web Software Enabled. web applications & APIs. finally these technologies led to modern Cloud Computing

Evolution Of Cloud Computing

Hardware Evolution
- Improved Computing power
- Better Servers
- Increased Storage capacity
- more efficient hardware

enabled →

Virtualization
↓
Virtualization
↓
Resource Sharing
↓
Better hardware Utilization

Supported →

Distributed System
↓
multiple Computers Connected through Networks
↓
Resource Sharing
↓
Distributed processing

Cloud Computing
- Iaas - Infrastructure as a Service.
- Paas - Platform as a Service
- Saas - Software as a Service

→

Service Delivery
↓
Computing resources delivered as Service
↓
on demand access
↓
pay-as-you-go-model.

Internet/web Software Evolution
↓
Internet Connectivity
↓
web application
↓
APIs
↓
Global access

Cloud Service Models

Cloud Service models describe the different levels of Computing resources and Services provided to users through the Cloud. They differ according to what the Provider manages and what the Customer is responsible for.

Iaas Stands for Infrastructure as a Service It provides Server, Storage, networking & VM's

Paas Stands for platform as a Service. It provides a platform for developing & running applications.

Saa's Stands for Software as a Service It provides ready to use Software applications through the Internet

Xaa's means Everything as a Service It provides manages the Infrastructure and Services assigned to it

The user manages applications and data According to the Service model.

Service Models.

Iaas

Infrastructure as a service

- VM, Storage, networking
- provides:
  - hardware
  - network
  - Storage
- users:
  - OS
  - application
  - data.

Paas

Platform layer

- Runtime, DB, Development

provides:

- hardware
- OS
- runtime

uses:

- applications
- data

Saas

Application layer

- Applications

provides

- Infrastructure
- platform
- Application

uses

data

Settings

users.

Xaas

Everything as a Service

- Database as Service
- Storage as Service
- function as a Service
- Security as a Service

Virtualization And Web Services As Enablers Of Cloud Computing

Virtualization and web Services are two important technologies that Support Cloud deployment and Scalability Virtualization allows Physical Computing resources to be divided into flexible Virtual environments.

- Virtualization is a technology creates VMs from Physical Computing resources.
- A hypervisor manages Real machines
- Virtualization provides abstraction between hardware and uses.
- It enables efficient resource Sharing and Improves hardware are resource utilization
- APIs provide acces to cloud Services and resource
- web Services Support Inseroperability between diffrent System
- SOAP and REST all used for web-based Communication.

Virtualization And Enablers Of Cloud

